Equipment Checkout System (ECS)

CEIS200 Software Engineering Project

Digitize inventory system to reduce lost/stolen tools and improve tracking

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PROJECT OVERVIEW

Core purpose: Replace the manual check-out and check-in process with a secure digital system to improve efficiency and accountability.

Goal: Create a clear digital audit trail for every tool transaction to prevent theft and mismanagement.

Scope: Web-based ECS with two user types (Standard, Administrative); MFA and RBAC for access control.

Data & UI: Centralized AWS database logging all transactions and equipment details; real-time history and an intuitive administrative dashboard.

PROJECT OVERVIEW

Core functions: manage check-out, check-in, inventory, and reporting.

Primary users: employees, administrators

High-level components: Web UI, Business Logic/API, Database, Authentication.

Key benefits: reduced losses, improved auditability, faster access, enhanced accountability.

Constraints: Budget limits; excludes integration with physical hardware such as scanners.

SOFTWARE REQUIREMENTS: FUNCTIONAL

User & Access Management: MFA, password resets, two user types (Standard, Administrative), RBAC.

Transaction Management: secure check-out/check-in, possession tracking, administrators set limits, audit logging.

Administrative Inventory Control: create/read/update/delete equipment and employee records.

SOFTWARE REQUIREMENTS: FUNCTIONAL

Data & Reporting: detailed logs for every transaction, historical views, and robust reporting.

Traceability: each functional requirement maps to test cases in the IEEE-829 test plan.

SOFTWARE REQUIREMENTS: NON-FUNCTIONAL

Performance: transactions complete within 2 minutes, search results within seconds, minimize data loss, scalability incorporated, redundancy included.

Security: MFA, encryption in transit and at rest, RBAC, audit logs.

Usability & Accessibility: intuitive UI with minimal training, clear error handling, low latency.

SOFTWARE REQUIREMENTS: NON-FUNCTIONAL

Reliability & Ethics: defined uptime targets, data accuracy, minimal collection of personal data.

Constraints & Assumptions: network availability, use of AWS cloud services, physical equipment not integrated during check-out or check-in, no artificial intelligence integration, $60,000 budget.

USE CASE MODELING

Actors: Standard Employee, Administrative Employee.

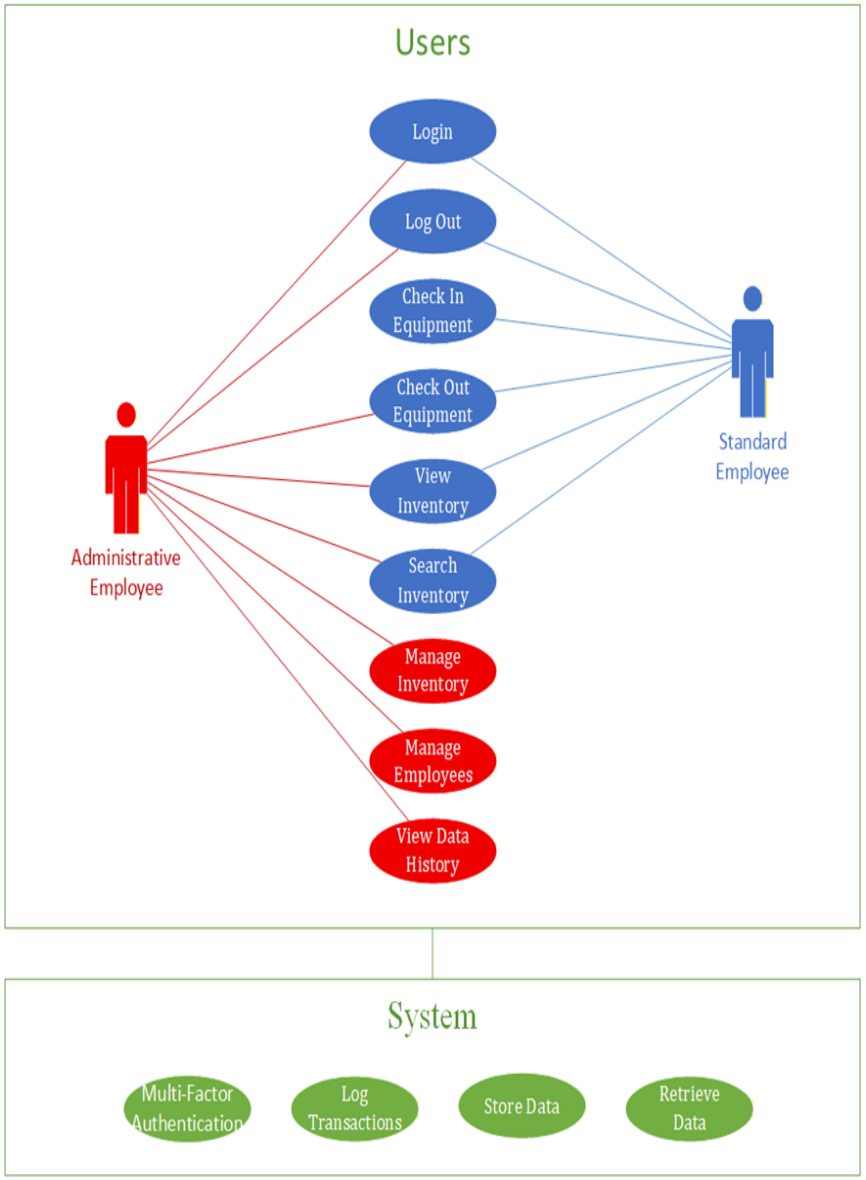
Standard Employee use cases: login/logout, checkout equipment, return equipment, view/search inventory.

Administrative Employee use cases: all Standard Employee actions plus manage inventory, manage employees, review historical data.

USE CASE MODELING

System boundary & shared services: MFA for secure access, transaction logging for auditability, centralized database storage/retrieval.

RBAC: roles separated; administrators have elevated privileges for management and auditing.



USE CASE MODELING

Use Case Diagram: shows actors, system boundary, and core use cases (Checkout, Check-in, Manage Inventory).

Class diagram summary: Primary classes: Equipment, Employee, Transaction with relationships (Transaction links Employee and Equipment; Equipment status attribute; Employee role attribute).

Architecture: client-server model using AWS database.

Traceability: map use cases to classes and to test cases in IEEE-829 test plan.

PROJECT PLANNING: WORK BREAKDOWN STRUCTURE SUMMARY

Seven main sections: Project Initiation & Planning; Design; Development; Testing; Reporting & Auditing; Deployment; Project Closure

Each section contains planning, execution, and verification activities

Focus areas: architecture & security, interfaces, inventory logic, access controls, test artifacts

Full WBS included in appendix

PROJECT PLANNING: GANTT CHART OVERVIEW

Timeline: 2025-12-01 → 2026-08-25 covering Initiation, Design, Development, Testing, Deployment, Closure

Key dependencies: design approval → development start; completion of testing → pilot rollout

Milestones tied to CEIS200 deliverables: Use Cases, Class Diagram, Test Plan, Test Cases

Appendix: full Gantt with task-level dates, durations, predecessors, and resource assignments

PROJECT PLANNING: SCHEDULE

Project start: 2025-12-01

Project end: 2026-08-25

Phases: Initiation & Planning → Design → Development → Testing → Reporting & Auditing → Deployment → Closure

Key deliverables aligned weekly with CEIS200 toolkit (requirements, design, tests, schedule)

Resource highlights: development team, QA, administrators, stakeholders

PROJECT PLANNING: DEPENDENCIES

Timeline summary: 2025-12-01 to 2026-08-25 (approx. 9 months)

Major milestones: Use Cases complete; Class Diagram complete; Test Plan drafted; Test Cases executed; Pilot rollout

Milestone sequencing: requirements → design → implementation → testing → pilot → deployment

Critical path emphasis: design approval and testing sign-off

OBJECT-ORIENTED ANALYSIS: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

Key objects: Equipment, CheckoutTransaction, UserAccount, InventoryItem, AccessControl, AuditLog

Core classes: User, EquipmentItem, CheckoutService, InventoryManager, RBACController, ReportGenerator

OBJECT-ORIENTED ANALYSIS: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

Responsibilities:

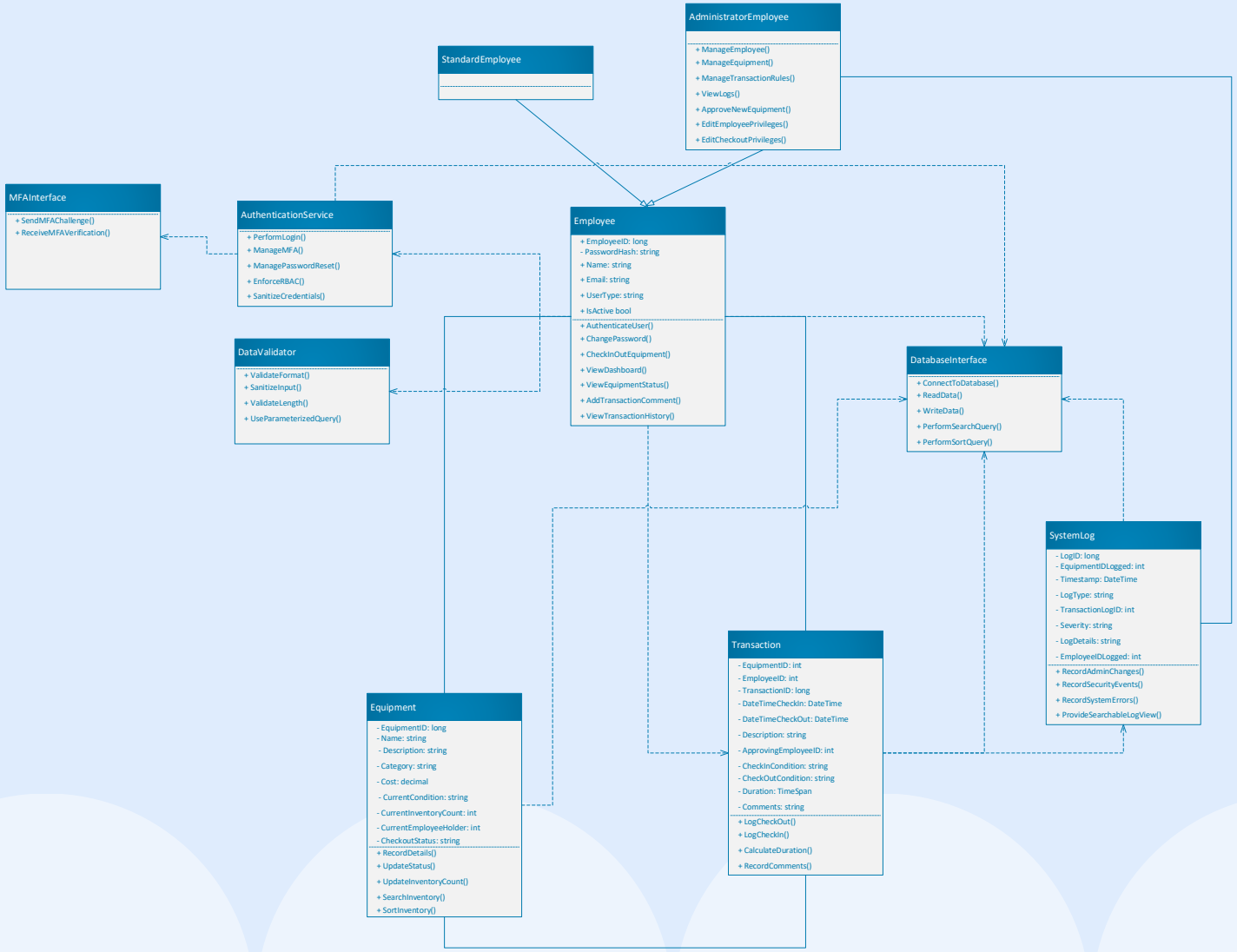
- User: identity, roles, authentication
- EquipmentItem: metadata, status, location
- CheckoutService: checkout/check-in logic, validation
- InventoryManager: counts, thresholds, notifications
- RBACController: enforce role-based permissions
- AuditLog: record actions, timestamps, user IDs

OBJECT-ORIENTED ANALYSIS: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

Relationships overview: Users interact with CheckoutService; CheckoutService updates InventoryManager and AuditLog; RBACController mediates access

Appendix: full UML class diagrams, per-class attributes/methods, and detailed relationships

OBJECT-ORIENTED ANALYSIS: CLASS DIAGRAM



TEST PLAN OVERVIEW (IEEE-829, CONCISE)

Purpose & Scope: Define QA strategy for ECS v1; validate MFA, transactions, RBAC, audit logging, inventory; non-functional checks (security, performance, data integrity)

Test approach: IEEE-829 based; unit, integration, system tests; traceability to requirements (RTM)

Representative test cases & outcomes:

ECS-TC-Checkout-01 — Pass (all steps passed; minor UI timing delay)

ECS-TC-INV-02-POS — Pass

ECS-TC-INV-02-NG — Pass (rejected invalid input; validation errors shown)

TESTING & SUMMARY REPORT OVERVIEW

Coverage & Criteria: Major functional requirements mapped to test cases; clear pass/fail criteria defined

Deliverables: Test logs, defect reports, test traceability matrix, test summary report

Appendix: full IEEE-829 test plan, detailed test cases, execution results, RTM, and logs

TESTING & SUMMARY REPORT: TEST DATA REALISM WITHOUT PII

Action: Used synthetic data, masked fields, and data templates. Also documented data handling rules.

Outcome: Testing proceeded safely with auditable data, while adhering to the ethical constraint of collecting only essential personal data.

TESTING & SUMMARY REPORT: TRACEABILITY ALIGNMENT

Action: Defined consistent IDs (e.g., REQ-4.3, REQ-4.4) and mapped them into test cases.

Outcome: Quality assurance through clear traceability established between requirements and tests.

TESTING & SUMMARY REPORT: BALANCING BREADTH VS. DEPTH

Action: Focused on three core functional tests plus negative and positive variants.

Outcome: Completed a solid foundation, noting gaps (RISK-003) for future targeted negative tests.

TESTING & SUMMARY REPORT: LESSONS LEARNED

Requirements: Clear, testable acceptance criteria prevent rework.

Design: Keep models focused on core use cases to reduce complexity.

Testing: Early traceability mapping saves time during integration.

Process: Frequent, short reviews reduced late-stage fixes.

TESTING & SUMMARY REPORT: CONCLUSION

The Equipment Checkout System (ECS) project successfully moved from a conceptual need to a fully defined, planned, and tested digital solution. The formal project artifacts (WBS, UML, Test Plan) are complete and confirm the system is technically ready for the final development and deployment phases, fulfilling the core objective of replacing the manual, error-prone process.

EXECUTIVE SUMMARY

Objective: Advance ECS from planning to test execution to reduce equipment loss and improve accountability.

Scope & Architecture: Web-based UI; RBAC for Standard Employees/Administrators; MFA via Microsoft Authenticator; AWS cloud database with real-time transaction & audit logging.

Key Deliverables: Defined functional & non-functional requirements; UML diagrams; IEEE-829 test plan with requirement traceability.

Current Status: Design complete; test plan drafted; test execution readiness established; focus on input sanitation, redundancy, and encryption.

EXECUTIVE SUMMARY

Risks & Mitigations: Data security, MFA adoption, risks mitigated via encryption, phased rollout, and strict test criteria.

Next Steps: Execute test plan (core functional & security cases), finalize deployment plan, begin pilot rollout.

Business Value: Expected reduction in equipment loss, improved accountability, and measurable cost savings.

Software Development Lifecycle (SDLC) Experience.

- Managed the project from end-to-end: Requirements > Planning > Design > Testing.
- Career Value: Demonstrates the ability to manage complex technical initiatives, not just perform isolated coding tasks.

CAREER READINESS

Structured Project Planning and Management

- Developed a definitive Work Breakdown Structure (WBS) and a Project Schedule (Gantt Chart).
- Career Value: Directly applicable to roles in Project Management and Team Leadership, showing competence in scope definition and resource planning.

CAREER READINESS

Security and Cloud Integration

- Designed the integration of external services: AWS Cloud (Database, Scalability, Redundancy) and Microsoft Authenticator (MFA, Security).
- Implemented core security features: RBAC (Role-Based Access Control) and Data Encryption.
- Career Value: This experience is vital for Cloud Engineers and DevOps Specialists, as it demonstrates practical skills in selecting, configuring, and securing critical third-party cloud services like AWS and integrated authentication providers to meet standards for scalability and data protection.

CAREER READINESS

Quality Assurance (QA) and Discipline

- Applied industry-standard methodology by creating an IEEE 829 Test Plan.
- Developed detailed positive, negative, and traceability-mapped test cases to ensure system quality and audit compliance.
- Career Value: Establishes a systematic approach to quality, vital for QA Engineers and highly valued in regulated technical environments.

CAREER READINESS

Design & Object-Oriented Analysis

- Translated business requirements into a structured system architecture using UML Class and Use Case Diagrams.
- Defined classes (e.g., Employee, Equipment) and core system logic.
- Career Value: The ability to translate abstract business needs into a structured system design using UML is a foundational skill for Software Architects and Senior Developers, demonstrating competence in building logical, scalable, and maintainable software.

CHALLENGE: SCHEDULING PHASES & BUDGET CONTROL

Challenge: Scheduling all phases efficiently while staying within the $60,000 budget.

Mitigation: Managed timeline and resource allocation using a Gantt chart; prioritized core features, tracked burn rate weekly, and reallocated tasks to keep costs on target.

CHALLENGE: MAINTAINING DATA INTEGRITY

Challenge: Data integrity risks from manual data entry.

Mitigation: Enforced input validation and standardized entry forms; implemented user compliance procedures, role-based data access, and detailed audit trails for error detection and correction.

CHALLENGE: SECURITY WITH THIRD-PARTY INTEGRATIONS

Challenge: Security concerns due to external integrations.

Mitigation: Use AWS database encryption (in transit and at rest), Microsoft Authenticator MFA, role-based access control, and comprehensive audit logging to trace and respond to incidents.

CHALLENGE: SCALABILITY

Challenge: System must scale as user load and inventory grow.

Mitigation: Deploy on AWS with tiered services and auto-scaling; plan capacity testing and staged scaling; design stateless services where possible to simplify scaling.

CHALLENGE: USER ADOPTION AND TRAINING

Challenge: Ensuring adoption with limited time for training.

Mitigation: Run a pilot rollout with one maintenance team, provide targeted hands-on training and quick reference guides, and collect feedback to iterate UI and workflows.

Appendix

APPENDIX: WORK BREAKDOWN STRUCTURE

1.0 Project Initiation & Planning

1.1.0 Project Definition & Authorization

1.1 Initiate Stakeholder Meeting

1.2 Define Project Charter (Scope, Requirements, Budget)

1.3 Define Project Scope Statement (Constraints, Assumptions, Deliverables)

1.2.0 Requirement Elicitation

1.2.1 Requirement Gathering & Analysis

1.2.2 Create Use Cases and User Stories

1.2.3 Conduct Vendor Meetings & Finalize Contracts (AWS & Microsoft)

APPENDIX: WORK BREAKDOWN STRUCTURE

1.3.0 Create Control Documents

1.3.1 Create Work Breakdown Structure (WBS)

1.3.2 Create Project Schedule

1.3.3 Develop Risk Management Plan

1.4 Conduct Approval Meeting with Stakeholders

APPENDIX: WORK BREAKDOWN STRUCTURE

2.0 Design

2.1.0 System Architecture & Security

2.1.1 Create Use Case Diagram

2.1.2 Create UML Class Diagram

2.1.3 Define System Architecture

2.1.4 Design AWS Database Structure and Schema

2.1.5 Define Security Architecture (MFA, Encryption, RBAC)

APPENDIX: WORK BREAKDOWN STRUCTURE

2.2.0 External System Integration Configuration

2.2.1 Design Cloud Environment and Redundancy (AWS configuration for scalability and reliability)

2.2.2 Define API Integration Flow (MFA, database and UI data flow)

2.3.0 Interface Design

2.3.1 Design Employee Dashboard User Interface (Quick Access Buttons, History, Comment Feed)

2.3.2 Design Administrator User Interface (User Management)

APPENDIX: WORK BREAKDOWN STRUCTURE

3.0 Development

3.1.0 Environment Setup & Security

3.1.1 Create AWS Database with Data Encryption

3.1.2 Establish Version Control Methods

3.1.3 Implement Role-Base Access Control (RBAS) System

3.1.4 Configure Data Redundancy and Backup Policies (AWS)

3.1.5 Create Multi-Factor Authentication (MFA) Workflow with Microsoft Authentication

APPENDIX: WORK BREAKDOWN STRUCTURE

3.2.0 Core Logic & Access Development

3.2.1 Create User Profiles (Administrator and Standard Employees)

3.2.2 Implement Password Reset Logic (Email Verification)

3.2.3 Implement Check-in / Check-out Logic with logging

3.3.0 Inventory & Administration Features

3.3.1 Implement Transaction Time Limit Restrictions

3.3.2 Implement Administrative Inventory Management (Add/Remove/Edit Equipment)

3.3.3 Implement Administrative User Management (Add/Remove/Edit Employees)

3.4.0 Interface and Data Handling

3.4.1 Develop User Interface and all required Input Fields

APPENDIX: WORK BREAKDOWN STRUCTURE

3.4.2 Implement Input Sanitation and Data Validation

(SQL injection prevention, character limits)

3.4.3 Develop Comment Area functionality (Notes, searchable thread)

3.5.0 Reporting & Auditing

3.5.1 Develop Reporting & History Views

(Employee & Equipment Transaction History)

3.5.2 Implementation of Search and Sort Logic for all Records

3.5.3 Create Structured Error and Administrative Logging

APPENDIX: WORK BREAKDOWN STRUCTURE

4.0 Testing

4.1 Functional testing to verify all features work

4.2 Performance testing to verify speed and scalability

4.3 Security Testing (Verify MFA, Encryption, RBAC)

4.4 User Acceptance Testing

APPENDIX: WORK BREAKDOWN STRUCTURE

5.0 Deployment

5.1 Stakeholder Training

5.2 Documentation Handover

5.3 System Installation and Launch

APPENDIX: WORK BREAKDOWN STRUCTURE

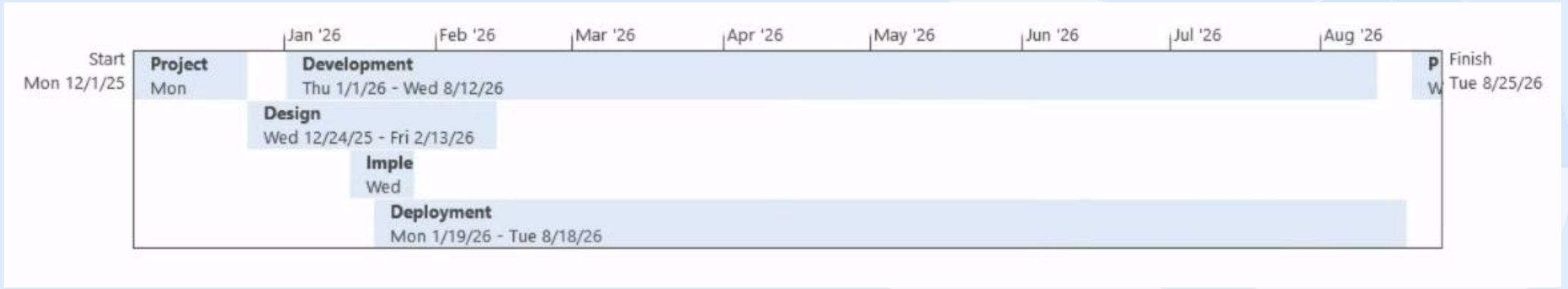
6.0 Project Closure

6.1 Formal Project Sign-off

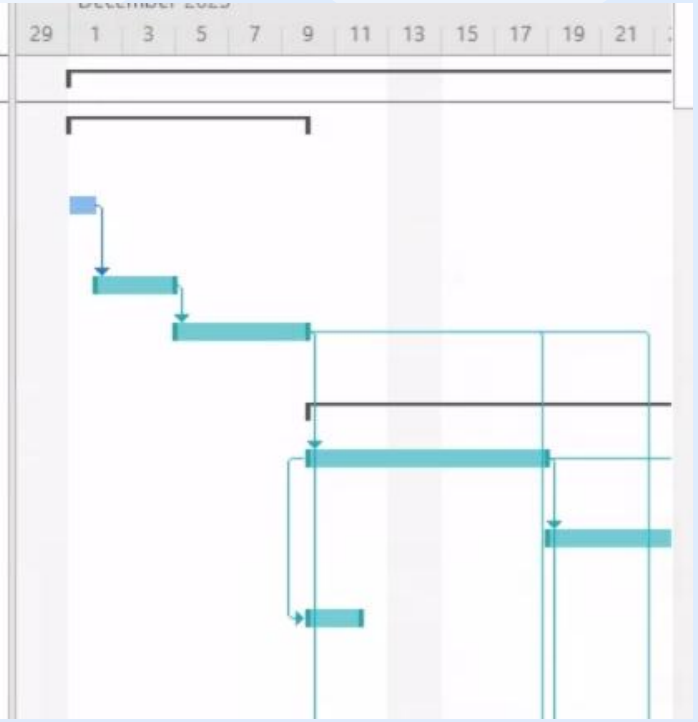
6.2 Verify Compliance (Delegate NIST and FTC requirements)

6.3 Archive Project

APPENDIX: GANTT CHART

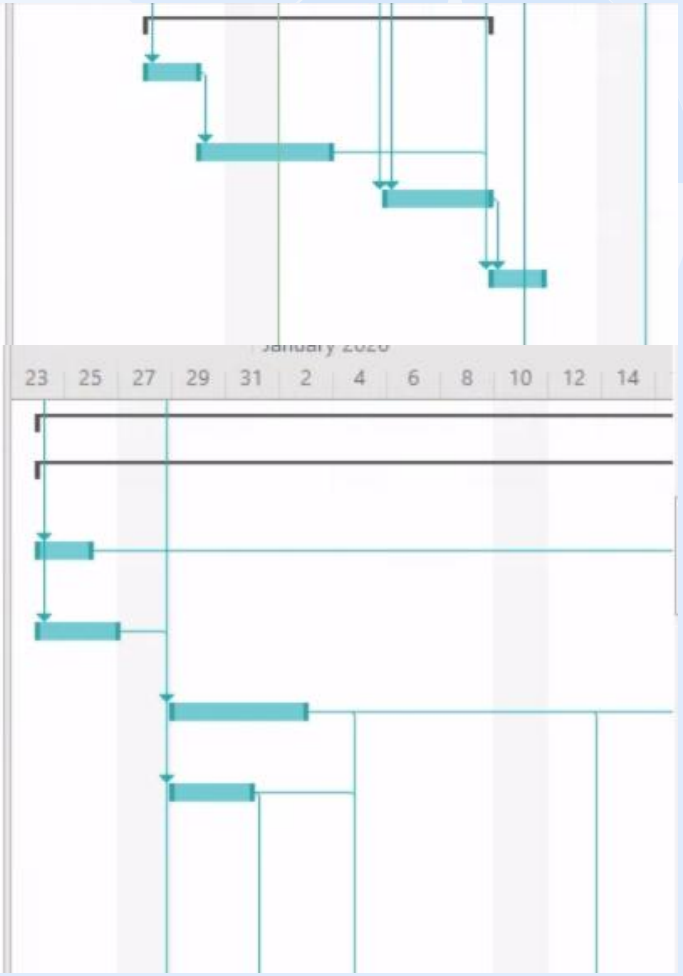


	Task Name	Duratio	Start	Finish	Predecessors
1	**Project Initiation & Planning**	**18 days**	**Mon 12/1/25**	**Wed 12/24/2**	
2	**Project Definition & Authorization**	**7 days**	**Mon 12/1/25**	**Tue 12/9/25**	
3	Initiate Stakeholder Meeting	1 day	Mon 12/1/25	Mon 12/1/25	
4	Define Project Charter	3 days	Tue 12/2/25	Thu 12/4/25	3
5	Define Project Scope Statement	3 days	Fri 12/5/25	Tue 12/9/25	4
6	**Requirement Elicitation**	**10 days**	**Wed 12/10/25**	**Tue 12/23/25**	
7	Requirement Gathering & Analysis	7 days	Wed 12/10/25	Thu 12/18/25	5
8	Create Use Cases and User Stories	3 days	Fri 12/19/25	Tue 12/23/25	7
9	Conduct Vendor Meetings & Finalize Contracts	2 days	Wed 12/10/25	Thu 12/11/25	7SS



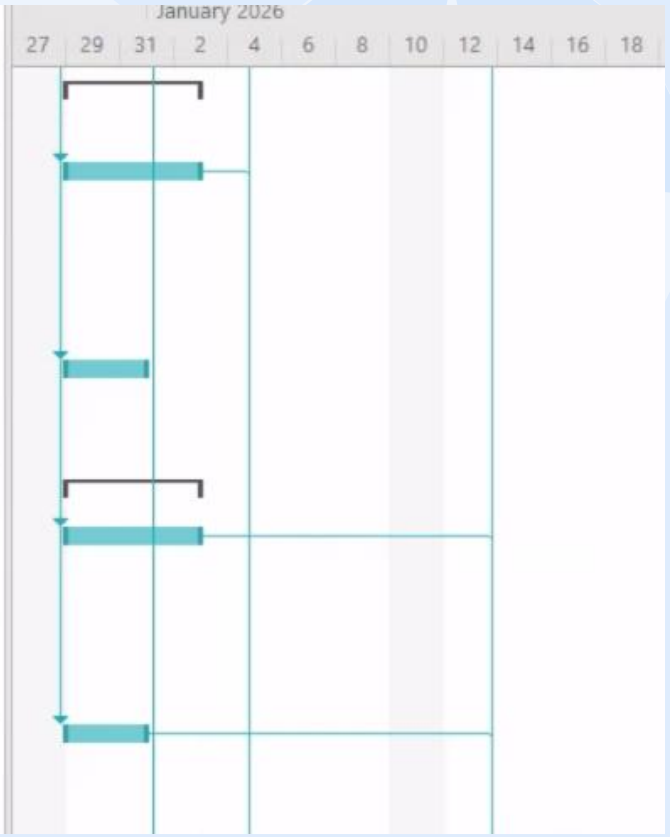
APPENDIX: GANTT CHART

		Task Name	Duration	Start	Finish	Predecessors
10		**Create Control Documents**	**9 days**	**Wed 12/10/25**	**Mon 12/22/2**	
11		Create Work Breakdown Structure (WBS)	2 days	Wed 12/10/25	Thu 12/11/25	5
12		Create Project Schedule	3 days	Fri 12/12/25	Tue 12/16/25	11
13		Develop Risk Management Plan	2 days	Fri 12/19/25	Mon 12/22/25	5,7
14		Conduct Approval Meeting with Stakeholders	2 days	Tue 12/23/25	Wed 12/24/25	5,12,13
15		**Design**	**38 days**	**Wed 12/24/25**	**Fri 2/13/26**	
16		**System Architecture & Security**	**38 days**	**Wed 12/24/25**	**Fri 2/13/26**	
17		Create Use Case Diagram	2 days	Wed 12/24/25	Thu 12/25/25	8
19		Define System Architecture	3 days	Wed 12/24/25	Fri 12/26/25	8
20		Design AWS Database Structure and Schema	5 days	Mon 12/29/25	Fri 1/2/26	7,19
21		Define Security Architecture (MFA, Encryption, RBAC)	3 days	Mon 12/29/25	Wed 12/31/25	7,19
18		Create UML Class Diagram	3 days	Wed 2/11/26	Fri 2/13/26	17,20



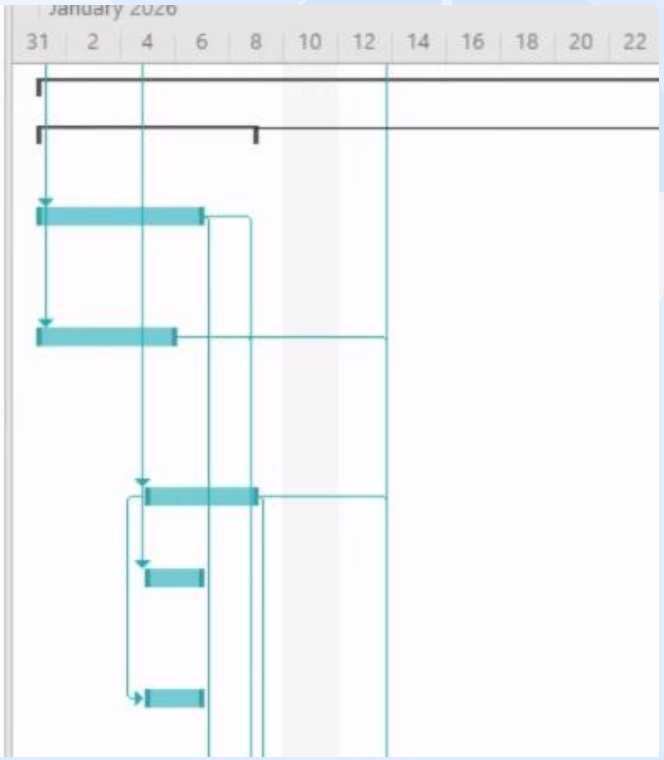
APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
22	**External System Integration Configuration**	**5 days**	**Mon 12/29/25**	**Fri 1/2/26**	
23	Design Cloud Environment and Redundancy (AWS configuration for scalability and reliability)	5 days	Mon 12/29/25	Fri 1/2/26	19
24	Define API Integration Flow (MFA, database and UI data flow)	3 days	Mon 12/29/25	Wed 12/31/25	7,19
25	**Interface Design**	**5 days**	**Mon 12/29/25**	**Fri 1/2/26**	
26	Design Employee Dashboard User Interface (Quick Access Buttons History, Comment Feed)	5 days	Mon 12/29/25	Fri 1/2/26	8,19
27	Design Administrator User Interface (User Management)	3 days	Mon 12/29/25	Wed 12/31/25	8,19



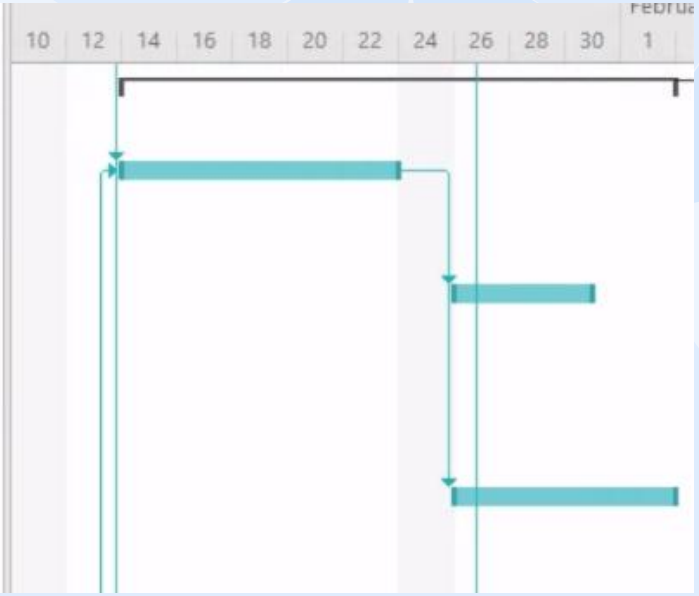
APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
28	**Development**	**160 days**	**Thu 1/1/26**	**Wed 8/12/26**	
29	**Environment Setup & Security**	**6 days**	**Thu 1/1/26**	**Thu 1/8/26**	
32	Implement Role-Base Access Control (RBAS) System	4 days	Thu 1/1/26	Tue 1/6/26	21
34	Create Multi-Factor Authentication (MFA) Workflow with Microsoft Authentication	3 days	Thu 1/1/26	Mon 1/5/26	21
30	Create AWS Database with Data Encryption	4 days	Mon 1/5/26	Thu 1/8/26	20,21
33	Configure Data Redundancy and Backup Policies (AWS)	2 days	Mon 1/5/26	Tue 1/6/26	23
31	Establish Version Control Methods	2 days	Mon 1/5/26	Tue 1/6/26	30SS

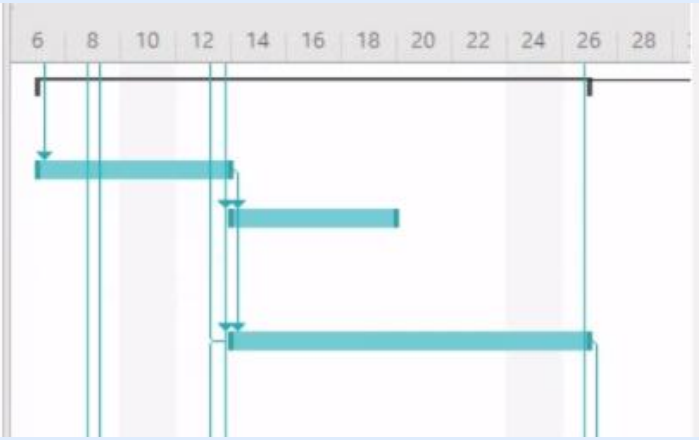


APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
43	**Interface and Data Handling**	**14 days**	**Wed 1/14/26**	**Mon 2/2/26**	
44	Develop User Interface and all required Input Fields	8 days	Wed 1/14/26	Fri 1/23/26	26,27,38SS
45	Implement Input Sanitation and Data Validation (SQL injection prevention, character limits)	5 days	Mon 1/26/26	Fri 1/30/26	44
46	Develop Comment Area functionality (Notes, searchable thread)	6 days	Mon 1/26/26	Mon 2/2/26	44

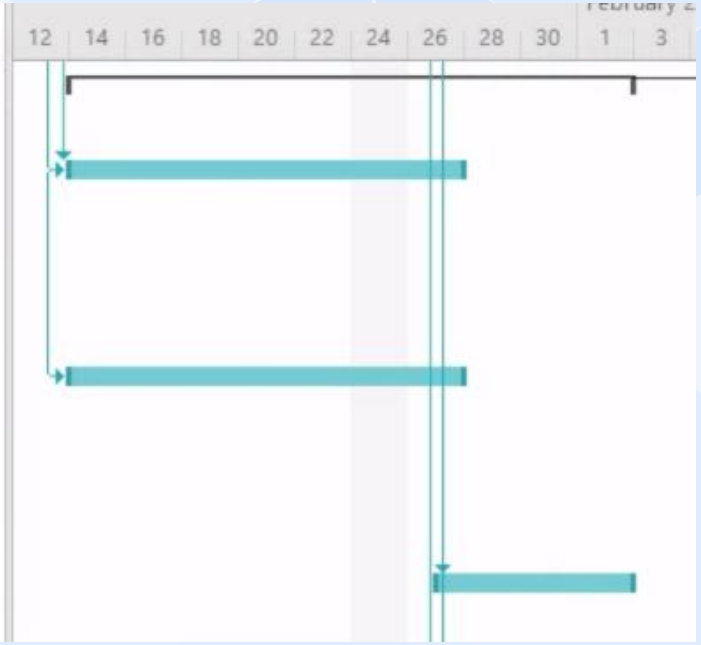


	Task Name	Duratio	Start	Finish	Predecessors
35	**Core Logic & Access Development**	**14 days**	**Wed 1/7/26**	**Mon 1/26/26**	
36	Create User Profiles	5 days	Wed 1/7/26	Tue 1/13/26	32
37	Implement Password Reset Logic (Email Verification)	4 days	Wed 1/14/26	Mon 1/19/26	34,36
38	Implement Check-in / Check-out Logic with logging	9 days	Wed 1/14/26	Mon 1/26/26	20,36



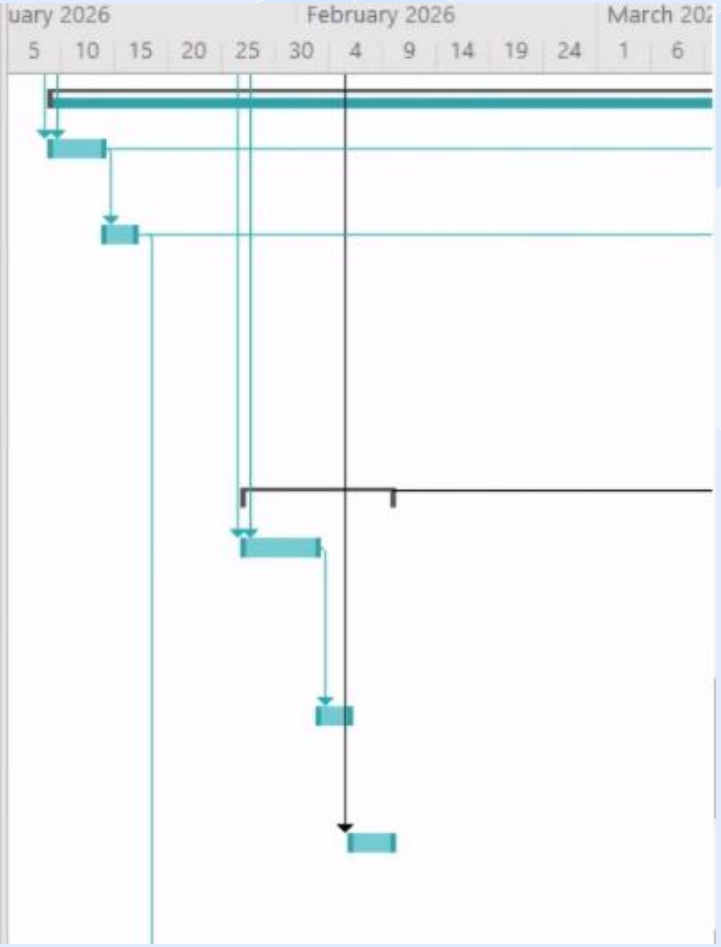
APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
39	**Inventory & Administration Features**	**14 days**	**Wed 1/14/26**	**Mon 2/2/26**	
41	Implement Administrative Inventory Management (Add/Remove/Edit Equipment)	10 days	Wed 1/14/26	Tue 1/27/26	27,30,38SS
42	Implement Administrative User Management (Add/Remove/Edit Employees)	10 days	Wed 1/14/26	Tue 1/27/26	41SS
40	Implement Transaction Time Limit Restrictions	5 days	Tue 1/27/26	Mon 2/2/26	38



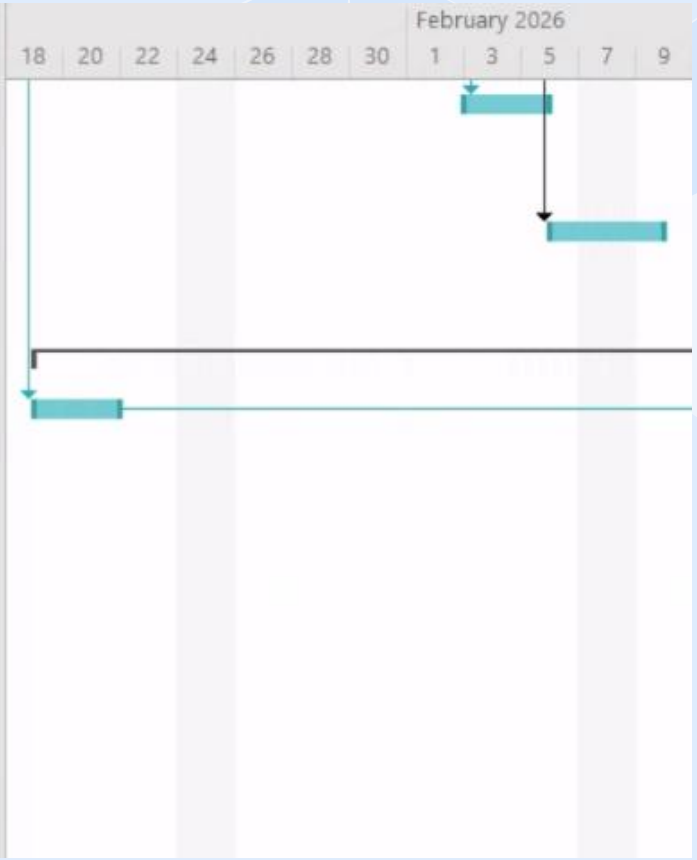
APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
51	**Testing**	**154 days**	**Fri 1/9/26**	**Wed 8/12/26**	
54	Security Testing (Verify MFA, Encryption, RBAC)	3 days	Fri 1/9/26	Tue 1/13/26	30,32,34
55	User Acceptance Testing	3 days	Wed 1/14/26	Fri 1/16/26	54
53	Performance testing to verify speed and scalability	3 days	Fri 7/31/26	Tue 8/4/26	54,55,29,35,39,43,47
52	Functional testing to verify all features work	6 days	Wed 8/5/26	Wed 8/12/26	29,35,39,43,47,54,53,55
47	**Reporting & Auditing**	**10 days**	**Tue 1/27/26**	**Mon 2/9/26**	
48	Develop Reporting & History Views (Employee & Equipment Transaction History)	5 days	Tue 1/27/26	Mon 2/2/26	20,38
49	Implementation of Search and Sort Logic for all Records	3 days	Tue 2/3/26	Thu 2/5/26	48
50	Create Structured Error and Administrative Logging	2 days	Fri 2/6/26	Mon 2/9/26	35,39



APPENDIX: GANTT CHART

	Task Name	Duratio	Start	Finish	Predecessors
49	Implementation of Search and Sort Logic for all Records	3 days	Tue 2/3/26	Thu 2/5/26	48
50	Create Structured Error and Administrative Logging	2 days	Fri 2/6/26	Mon 2/9/26	35,39
56	**Deployment**	**152 days**	**Mon 1/19/26**	**Tue 8/18/26**	
57	Stakeholder Training	3 days	Mon 1/19/26	Wed 1/21/26	55
58	Documentation Handover	2 days	Thu 8/13/26	Fri 8/14/26	28,51
59	System Installation and Launch	2 days	Mon 8/17/26	Tue 8/18/26	55,57,58
60	**Project Closure**	**5 days**	**Wed 8/19/26**	**Tue 8/25/26**	
61	Formal Project Sign-off	2 days	Wed 8/19/26	Thu 8/20/26	59
62	Verify Compliance (Delegate NIST and FTC requirements)	2 days	Fri 8/21/26	Mon 8/24/26	61
63	Archive Project	1 day	Tue 8/25/26	Tue 8/25/26	62,61



APPENDIX: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

Employee

EmployeeID, PasswordHash, Name, Email, UserType, IsActive, AuthenticateUser(), ChangePassword(), CheckInOutEquipment(), ViewDashboard(), ViewEquipmentStatus(), AddTransactionComment(), ViewTransactionHistory()

StandardEmployee

Inherits from Employee

AdministratorEmployee

Inherits from Employee, ManageEmployee(), ManageEquipment(), ManageTransactionRules(), ViewLogs(), ApproveNewEquipment(), EditEmployeePrivileges(), EditCheckoutPrivileges()

APPENDIX: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

Equipment

EquipmentID, Name, Description, Category, Cost, CurrentCondition, CurrentInventoryCount, CurrentEmployeeHolder, CheckoutStatus, RecordDetails(), UpdateStatus(), UpdateInventoryCount(), SearchInventory(), SortInventory()

Transaction

TransactionID, EquipmentID, EmployeeID, DateTimeCheckIn, DateTimeCheckOut, Description, ApprovingEmployeeID, CheckInCondition, CheckOutCondition, Duration, Comments, LogCheckOut(), LogCheckIn(), CalculateDuration(), RecordComments()

APPENDIX: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

SystemLog

LogID, EquipmentIDLogged, Timestamp, LogType, TransactionLogID, Severity, LogDetails, EmployeeIDLogged, RecordAdminChanges(), RecordSecurityEvents(), RecordSystemErrors(), ProvideSearchableLogView()

AuthenticationService

PerformLogin(), ManageMFA(), ManagePasswordReset(), EnforceRBAC(), SanitizeCredentials()

DatabaseInterface

ConnectToDatabase(), ReadData(), WriteData(), PerformSearchQuery(), PerformSortQuery()

APPENDIX: OBJECT IDENTIFICATION & CLASS RESPONSIBILITIES

MFAInterface

SendMFAChallenge(), ReceiveMFAVerification()

DataValidator

ValidateFormat(), SanitizeInput(), ValidateLength(), UseParameterizedQuery()

APPENDIX: TEST PLAN (IEEE-829, CONCISE)

Test Case: Checkout - Standard Employee

Test Case ID: ECS-TC-Checkout-01

Test Item Description: Successful Standard Employee Equipment Checkout

Status: Not Executed

Pre-conditions: User logged in as Standard Employee; MFA completed; Item available (inventory > 0)

Requirements Mapped: REQ-Login, REQ-Checkout, REQ-TransactionLogging

Steps: Navigate to checkout page → Search/select item → Enter required checkout details → Click Checkout → Confirm checkout

Expected Results: Login and MFA pass; item quantity > 0; checkout confirms; transaction logged; inventory -1

Post-conditions: Inventory decreases by 1; audit log updated

Traceability: REQ-4.3, REQ-4.4

Notes: Verify DB rows in Transaction and Inventory

APPENDIX: TEST PLAN (IEEE-829, CONCISE)

Test Case: Inventory - Add New Equipment (Negative)

Test Case ID: ECS-TC-INV-02-NG

Test Item Description: Negative case for adding equipment with invalid data

Status: Not Executed

Pre-conditions: Admin login; MFA; AWS DB accessible

Steps: Login as Admin → Inventory Management → Add New Equipment → Enter invalid data → Submit and verify validation errors

Expected Results: Validation errors shown; no record created; no audit log

Post-conditions: No new equipment; audit unchanged

Traceability: REQ-AdminPrivileges, REQ-EquipmentRecord, REQ-EmployeeLog, REQ-EquipmentLog

Notes: Include data sanitation checks

APPENDIX: TEST PLAN (IEEE-829, CONCISE)

Test Case: Inventory - Add New Equipment (Positive)

Test Case ID: ECS-TC-INV-02-POS

Test Item Description: Positive case for adding equipment with valid data

Status: Not Executed

Pre-conditions: Admin login; MFA; AWS DB accessible

Requirements Mapped: REQ-AdminPrivileges, REQ-EquipmentRecord, REQ-EmployeeLog

Steps: Login as Admin → Inventory Management → Add New Equipment → Enter valid data → Submit and verify success → Verify DB record and audit log

Expected Results: Admin dashboard; record saved; audit log updated

Post-conditions: New equipment exists; audit updated

Traceability: REQ-AdminPrivileges, REQ-EquipmentRecord, REQ-EmployeeLog, REQ-EquipmentLog

Notes: Ensure field validation passes and audit integrity

Equipment Checkout System (ECS)

CEIS200 Software Engineering Project

Digitize inventory system to reduce lost/stolen tools and improve tracking

Francisco Acosta

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